Earth-State Embeddings:
a self-supervised codec of the North Atlantic, its transport read-out, and what a transformer forecasts from it

blauewelt/earth — technical report

2 September 2026

Abstract

We compress the observed state of each North Atlantic grid cell — ocean reanalysis, Argo climatology, wind stress, sea-surface temperature — into a 32-dimensional embedding with a self-supervised per-pixel codec, and train a transformer to predict the embedding forward at five-day cadence. **Codec.** Read out by an attention head over the individual pixels of the 26.5°N section, the frozen embeddings predict RAPID overturning transport at $r = 0.67$ (two codec seeds, year-blocked k-fold); the same head fed raw section values reaches 0.66, so the representation adds nothing measurable for the current-state read-out. Wind stress accounts for most monthly skill (a wind-only ridge reaches 0.53); the 16 Argo levels move skill to the density-driven MOVE transport at 16°N; neither codec capacity (0.92M to 40.7M) nor training length (50k to 1M steps) moves the probe. **Forecaster.** Under a training pool that excludes every window touching a held-out year, stage-2 heads from 7.6M to 400M parameters reach their best held-out one-step loss inside the first ∼2,000 of 200,000 steps and worsen thereafter, the best level lying in 0.58–0.62 of the persistence error across a 53× span in capacity. Rolled twelve months from held-out starts, the 206.7M head's field anomaly correlation decays from 0.61 at five days to 0.41 at ten, ≈ 0.1 from two months on and −0.03 at a year; it beats raw persistence at every lead but the last and damped persistence at none. Its squared-error score is negative (mean MSSS −0.44 against climatology) because it emits anomalies at 78% amplitude at 10% correlation — an identity reproduces all 73 leads to 0.0135, and the same states rescaled would score ≈ +0.02. Skill that survives past a few pentads lives in sea-surface temperature; 32 of 40 channels score nothing at any lead. A 200-mode linear inverse model fitted to the same field and rolled from the same starts is less correlated than the head at five and ten days and more correlated at every lead from fifteen days to a year (0.26 at 90 days, 0.18 at 365), with a squared-error score that stays positive throughout. A 7.6M head rolled through the same battery has the same mean correlation as the 206.7M head (0.103 against 0.105) and does not decay through zero, holding ≈ 0.11 from two months to a year; its squared-error score is less negative (−0.18) only because it emits smaller anomalies. Transport at 26.5°N shows no measurable skill at roughly nine effective starts. Every rolled number is a single seed.

1 Data

The working tensor covers 0–70°N, 100°W–20°E on a 0.25° grid (281 × 481 cells, 86,698 ocean) at five-day cadence from 1982-01 to 2024-12 (3,142 bins), with 40 channels per cell: surface current speed, log mixed-layer depth and sea-surface height from the GLORYS reanalysis; temperature and salinity at 16 pressure levels from the Roemmich–Gilson Argo climatology (native 1°, monthly, from 2004); NCEP Reanalysis-1 wind stress and its within-bin variability (four channels); OISST sea-surface temperature. Channels absent before their record begins enter as missing tokens. Earlier codec generations used 1° monthly tensors with 12–25 channels, on the same window or globally; they are identified by channel count below. The transport truth is the RAPID-MOCHA overturning at 26.5°N (2004–), deseasonalised; MOVE at 16°N is a second target. Everything cited here — tensors, weights, result bundles — is public (Appendix A).

2 The codec and its read-outs

Model. A masked autoencoder over one pixel's channels at one time bin produces a d_z-dimensional embedding; a 3 × 3-patch variant sees the pixel's neighbours. The working codec on the 0.25° pentad

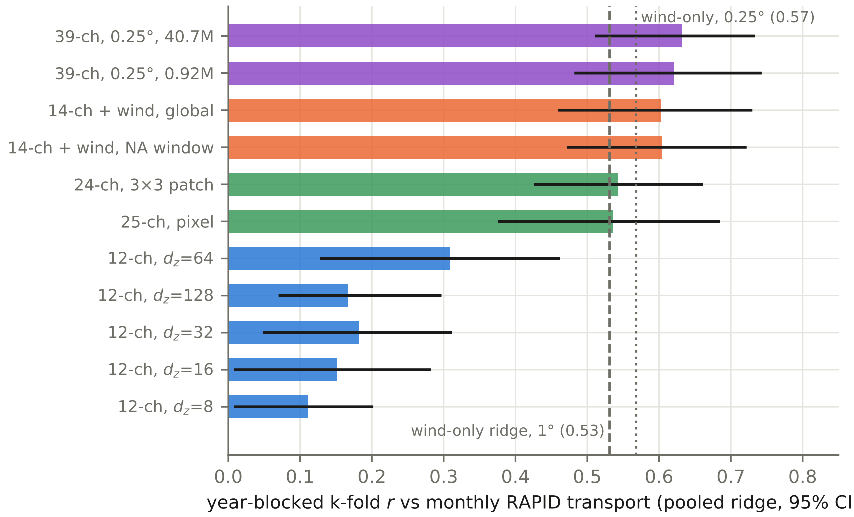

Figure 1: Pooled k-fold correlation with monthly deseasonalised RAPID transport by codec generation. Blue: the 12-channel width sweep; orange: with wind stress; green: 24–25 channels with 8 Argo levels; violet: the 0.25° tensor with 16 levels and SSH at two capacities.

tensor is 512-wide, 12 layers, decoder width 256, d_z=32, patch 1, 37.976M parameters, trained 200k steps at batch 512 with 2009, 2017 and 2023 held out of training and of the anomaly statistics. Embeddings are probed for transport with a year-blocked k-fold: a ridge on the section-mean embedding (*pooled*), or an attention head over the section's individual pixels (*unpooled*). Field reconstruction is scored as the percentage improvement over persistence, one bin ahead.

What moves the transport read-out. Figure 1 is the pooled ranking across generations. Adding wind stress to a 12-channel tensor takes the pooled probe from 0.31 to 0.60; a wind-only ridge with no embedding reaches 0.531 on the 1° tensors and 0.568 on the 0.25° one, so most monthly skill at 26.5°N is Ekman physics. Training on the global ocean instead of the window costs the Atlantic read-out nothing (0.602 against 0.604). Bottleneck width matters little for reconstruction (saturating by d_z=16–32) and peaks at d_z=64 for the probe, turning over at 128 against ∼220 truth months per fold. On the 0.25° tensor a 0.92M pilot reads 0.620 and the 40.7M codec 0.631 — 44× the parameters buys +0.011, inside seed noise (±0.05 on the pooled probe) — and a codec trained from 50k to one million steps is flat from the first probe to the last. Neither capacity nor training length is what stands between this codec and a better transport number.

Pooling was hiding the signal, and the codec is not what recovers it. The same frozen embeddings read three ways (Figure 2): an MLP on the section mean does no better than the ridge, so the ridge was never the limit as a function class; the attention head over the unpooled section beats every pooled probe and lifts the 3 × 3-patch codec to 0.690 [0.570, 0.781] (RMSE 2.02 Sv against a 2.79 Sv target). Geostrophic transport is the east–west contrast across the section, and a section mean is the statistic in which that contrast cancels. The same head fed raw section anomalies instead of embeddings completes the attribution:

tokens carry	raw data	codec embedding
single pixel	0.613 [0.48, 0.73]	0.617 [0.49, 0.73]
3 × 3 neighbourhood	0.659 [0.55, 0.75]	0.690 / 0.654 (two codec seeds)

Unpooling the section is the whole of the improvement (pooled 0.54 → head 0.65+); the 3 × 3 receptive field adds ≈ +0.045 and does so on raw data alone; pretraining adds +0.013 at two seeds, which is nothing against a head seed spread of 0.036. On the 0.25° monthly tensor the margin is +0.034 (0.662 against 0.628, one seed). At pentad cadence (n=1,459 transport bins) three unpooled read-outs — the frozen monthly codec (0.691), a codec trained on the pentad tensor itself (0.680), and

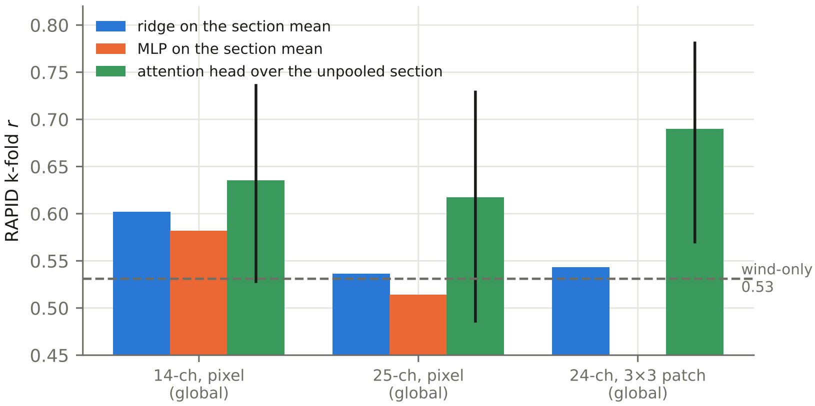

Figure 2: Same frozen embeddings, same folds, three read-outs. Head bars carry 95% intervals.

raw fields with no codec (0.683) — land within 0.011 of one another, while the pooled ridge (0.660) sits below the pentad wind bar (0.670). For the current-state transport read-out, read-out design and not representation learning is what moved the number.

Where the deep channels go. Extending Argo from 4 to 16 levels did not raise the RAPID probe but roughly doubled skill at MOVE — a deep, density-driven transport — from 0.111 to 0.206 [0.044, 0.346], the first multi-target result whose interval excludes zero, with the 18-month low-passed correlation rising from 0.38 to 0.62. At 16°N the wind-only ridge is negative (-0.376). One task-blind representation serves a wind-dominated and a density-dominated target when the read-out respects the structure of the question.

Event anatomy. Out of fold, the winter 2009–10 collapse — the sharpest event in the RAPID record — is captured at 16% of its amplitude without wind channels, 50% with them, 59% by the 25-channel codec and 51% by the 40.7M 0.25° codec. Average correlation and event capture are different capabilities; the 25-channel codec has the best dip and the second worst monthly r.

Quantizing the embedding. A finite-scalar-quantized bottleneck (FSQ: each of d_z=6 dimensions rounded to 8, 8, 8, 5, 5 or 5 levels, one 2^{16}-way code per pixel and bin, a LayerNorm bound on the pre-quantization activation) trained from a random initialisation collapses on the pentad tensor: the encoder becomes input-independent within a few thousand steps (fitted level-spread 0.638 → 0.005 by step 7,500, twice), and an unbounded 8-level lattice on d_z=32 drifts to pre-quantization magnitudes $\sim 10^4$ against a lattice radius of 2, wearing its eight levels as a one-bit sign code. Switching the same lattice on at step 200k of a finished continuous d_z=6 codec and training 60k further steps keeps the activations input-dependent throughout (level-spread 0.97 → 0.80), at an 18% reconstruction tax (0.270 against 0.229) and a per-channel one-step error of 0.742 against 0.681 (persistence 0.843). The quantized codec reads RAPID at 0.588 [0.534, 0.641] against the continuous 0.579 [0.525, 0.627] through the unpooled head, one seed each — indistinguishable, and both below the raw 3×3 control at this width (0.693).

3 The forecaster

Model and protocol. A causal transformer over a window of K=144 consecutive embeddings at one pixel (two years) plus a 145-point spatial stencil on a spiral ring predicts the embedding one bin ahead, with input noise at 0.7 of the persistence scale, gradient clipping at 128, batch 256, learning rate 10^{-3} decaying exponentially (half-life 40k steps, warm-up 2k). The loss is dense over all 144 window positions, so the pool admits only windows that touch no held-out bin anywhere the forward pass can see (209,549,066 windows: 2,417 end-times × 86,698 pixels); it certifies this by brute force before training. The held-out one-step loss is the z-space MSE relative to persistence on held-out windows (denominator 21.44621). Widths: 256×8 (7.598M), 512×12 (40.388M), 1024×16 (206.659M), 1280×20 (399.948M); all on the frozen 37.976M codec and its embedding of the tensor.

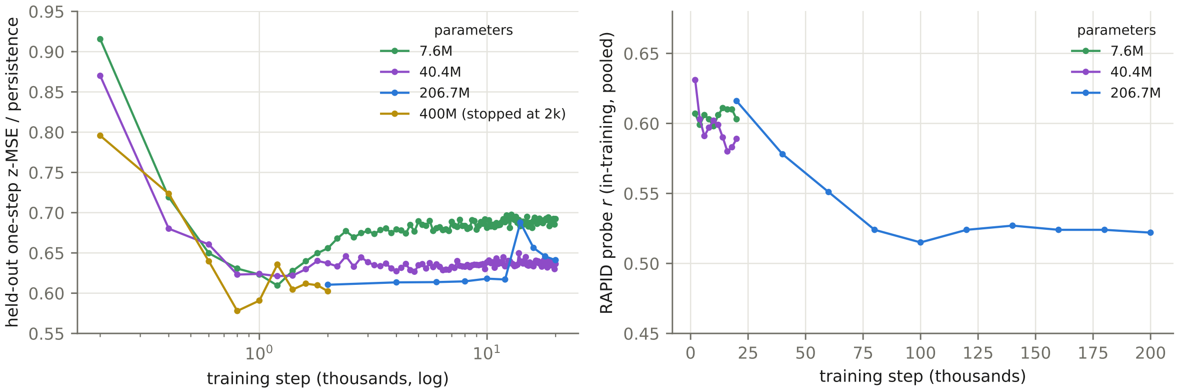

Figure 3: Left: held-out one-step ratio over the first 20,000 steps, by width. Right: the in-training RAPID probe over the full run.

params	best held-out ratio	at step	ratio at 20k	ratio at end	train ratio at end
7.598M	**0.6095**	1,200	0.692	0.692 (20k)	0.146
40.388M	0.6212	1,200	0.636	0.636 (20k)	0.086
206.659M	0.6105	$\leq$ 2,000	0.641	0.686 (200k)	0.023
399.948M	0.5779	800	—	0.602 (2k)	0.171

Table 1: Held-out one-step ratio by width. The 206.659M run is recorded every 2,000 steps, so its minimum is at or before its first record; the others, recorded every 200, bottom out at 1,200.

A *roll* advances every ocean pixel one bin at a time from a true 144-bin context, feeding predictions back, for 73 steps (365 days), decodes through the codec and scores standardised anomalies per pixel and channel on three scopes: gate (600 random pixels plus the section), corridor (the fastest quarter of the window by current speed, dilated two cells, plus the section; 30,158 pixels), window (all pixels). Starts are three per held-out year, nine in all. Per lead: $\mathrm{MSSS}_x = 1 - \mathrm{MSE}/\mathrm{MSE}_x$ against climatology (zero anomaly), raw persistence, and damped persistence (per-pixel AR(1) fitted on training years); anomaly correlation ACC; amplitude ratio a. The transport read-out regresses the rolled section states onto RAPID and reports r in three lead bands.

The held-out minimum is early at every width. Figure 3 and Table 1. All four widths reach their best held-out ratio inside 2,000 steps and get worse afterwards; the best levels span 0.578–0.621 across 53× in parameters. At step 20,000 the 7.598M arm (0.692) is 0.051 worse than the 206.659M arm (0.641), and the ordering is not monotone in width (40.388M reads 0.636). Width buys how fast an arm reaches its minimum and how badly it then degrades, not the level of the minimum. The in-training transport probe separates the arms the other way: 7.598M holds 0.598–0.611 across ten probes, 40.388M drifts to 0.589, and 206.659M falls from 0.616 at 20k to 0.522 at 200k. The training loss keeps falling throughout (to a ratio of 0.023 at 200k for the 206.659M head, a 30× train/held-out gap): the model memorises the training years long after it has learned anything that transfers.

The twelve-month roll. The 206.659M head at its 200k end state, rolled from the nine held-out starts (Table 2, Figure 4). Corridor anomaly correlation is 0.606 at five days, 0.412 at ten, 0.265 at fifteen, 0.20 at a month, near 0.1 from two months on, −0.03 at a year. The head beats raw persistence at 72 of 73 leads (mean $\mathrm{MSSS}_{\mathrm{pers}}$ +0.204) and damped persistence at none: $\mathrm{MSSS}_{\mathrm{damped}}$ runs from −0.011 at five days to −0.719 at a year (mean −0.461; gate −0.416, window −0.420). A per-pixel AR(1) decay toward climatology, fitted on training years, is a better forecast of this field than the transformer at every horizon.

The negative score is amplitude, not sign. For a forecast with correlation ACC and amplitude ratio a against a zero-anomaly reference, $\mathrm{MSSS}_{\mathrm{clim}} = 1 - (1 + a^2 - 2a\,\mathrm{ACC})$ identically. From the head's own per-lead ACC and a this reproduces the 73 measured values to a mean absolute error of 0.0135

	206.659M head					LIM, 200 modes		
lead (days)	$\text{MSSS}_{\text{clim}}$	$\text{MSSS}_{\text{pers}}$	$\text{MSSS}_{\text{damped}}$	ACC	a	$\text{MSSS}_{\text{clim}}$	$\text{MSSS}_{\text{damped}}$	ACC
5	+0.365	+0.194	−0.011	0.606	0.692	+0.169	−0.323	0.406
10	+0.109	+0.252	−0.131	0.412	0.673	+0.135	−0.097	0.361
15	−0.172	+0.178	−0.354	0.265	0.761	+0.106	−0.034	0.324
30	−0.168	+0.252	−0.244	0.204	0.672	+0.069	+0.008	0.258
60	−0.320	+0.201	−0.342	0.107	0.690	+0.056	+0.040	0.227
90	−0.322	+0.211	−0.333	0.132	0.725	+0.075	+0.067	0.261
180	−0.354	+0.337	−0.357	0.153	0.778	+0.053	+0.051	0.223
270	−0.479	+0.227	−0.479	0.009	0.720	+0.034	+0.034	0.191
365	−0.719	−0.101	−0.719	−0.031	0.822	+0.019	+0.019	0.177
mean of 73 leads	−0.439	+0.204	−0.461	0.105	0.780	+0.054	+0.034	0.225

Table 2: The 206.659M head and the 200-mode linear inverse model, rolled from the same nine held-out starts, corridor scope. Head, gate and window scopes: mean $\text{MSSS}_{\text{clim}}$ −0.395 and −0.401, mean ACC 0.131 and 0.121; LIM: +0.061 and +0.053, 0.253 and 0.233. The LIM's amplitude ratio falls from 0.48 at five days to 0.06 at a year. Single seed for the head.

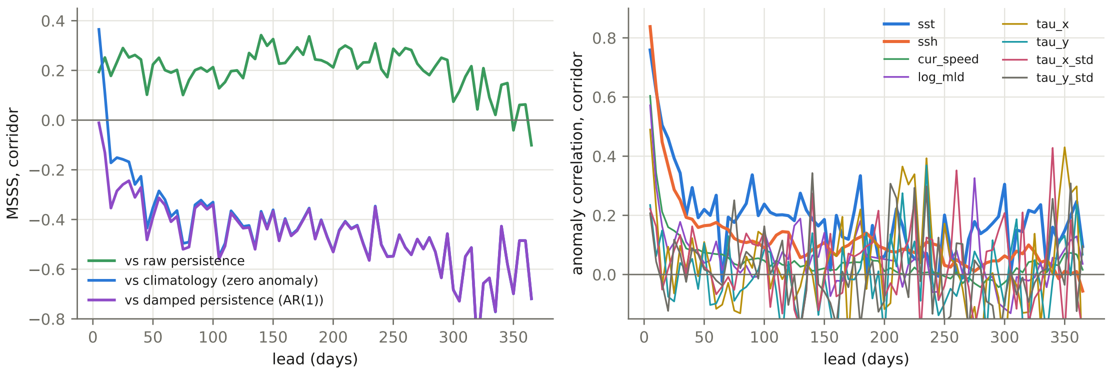

Figure 4: Left: the rolled head against its three references, corridor scope. Right: anomaly correlation against lead per scoreable channel; the 32 Argo channels have no scored samples at any lead.

(Figure 5). The head emits anomalies at $a \approx 0.78$ while its correlation is 0.1; rescaled to $a = \text{ACC}$ the same states would score ACC^2 per lead, a corridor mean of about +0.02. Whether a calibration fitted on training-year starts transfers to held-out starts is untested.

A linear inverse model. A linear inverse model fitted to the same standardised anomaly field — the eight scoreable channels over the window's 86,698 ocean pixels, reduced by PCA to K modes, with the propagator $\mathbf{G}(\tau) = \mathbf{C}(\tau)\mathbf{C}(0)^{-1}$ estimated at τ of one pentad from consecutive training bins only — is rolled from the same nine starts and scored through the same battery (Table 2, Figure 5). At 200 modes (76% of the training variance; spectral radius 0.986, leading e-folding time 360 days) its corridor correlation is 0.41 at five days, 0.26 at 90 and 0.18 at 365; its amplitude falls with lead in step with its correlation, so $\text{MSSS}_{\text{clim}}$ stays positive at every lead (+0.17 at five days, mean +0.054). The head's correlation exceeds the LIM's at five and ten days only; from fifteen days on the LIM is more correlated with the truth at every lead, and its mean squared-error score is above both the head's measured value and the head's amplitude-calibrated value (+0.019). Against damped persistence the LIM loses at leads up to 15 days and wins by 0.01–0.07 from 30 days out; the head loses at every lead. Fifty and 100 modes score below 200 at every lead (mean correlation 0.200, 0.209, 0.225). Per channel, the LIM holds sea-surface temperature at 0.48, 0.41, 0.33 and 0.28 at 5, 30, 90 and 365 days against the head's 0.76, 0.34, 0.34 and 0.10, and sea-surface height at 0.57, 0.36, 0.30, 0.30 against 0.84, 0.25, 0.11, −0.06. The LIM's training set is every bin outside a held-out year (2,923 bins, 2,919 pairs), which is slightly larger than the pool the head's 144-bin windows admit.

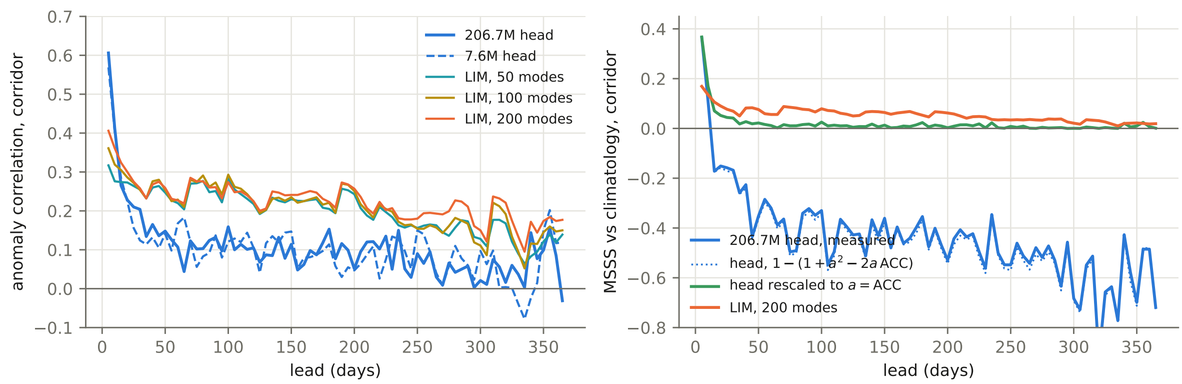

Figure 5: Left: corridor anomaly correlation per lead for the 206.659M and 7.598M heads and linear inverse models with 50, 100 and 200 modes, from the same starts. Right: the head's measured $\mathrm{MSSS}_{\mathrm{clim}}$, the value the identity predicts from its own ACC and a, the value the same states would score at $a = \mathrm{ACC}$, and the 200-mode LIM.

	ACC		$\mathrm{MSSS}_{\mathrm{clim}}$		a	
lead (days)	7.598M	206.659M	7.598M	206.659M	7.598M	206.659M
5	0.569	0.606	+0.314	+0.365	0.69	0.69
10	0.403	0.412	+0.125	+0.109	0.61	0.67
15	0.284	0.265	−0.005	−0.172	0.58	0.76
30	0.124	0.204	−0.160	−0.168	0.55	0.67
90	0.137	0.132	−0.120	−0.322	0.52	0.72
180	0.114	0.153	−0.139	−0.354	0.51	0.78
365	0.119	−0.031	−0.181	−0.719	0.54	0.82
mean of 73 leads	0.103	0.105	−0.179	−0.439	0.54	0.78

Table 3: The 7.598M and 206.659M heads through the same battery, corridor scope. Mean $\mathrm{MSSS}_{\mathrm{damped}}$ −0.200 and −0.461; gate and window mean $\mathrm{MSSS}_{\mathrm{clim}}$ −0.201/−0.213 and −0.395/−0.401. Single seed each.

Width on the rolled field. The 7.598M head (256 × 8, K= 144, its 20,000-step end state) rolled through the identical battery scores the same field as the 206.659M head (Table 3, Figure 5): mean corridor correlation 0.103 against 0.105, and the same amplitude-calibrated ceiling (mean ACC^2 0.0185 against 0.0188). The two differ at the ends of the lead axis. The larger head is more correlated over the first ten days (0.606 and 0.412 against 0.569 and 0.403); the smaller one does not decay through zero, holding 0.10–0.14 from 60 days to a year where the larger falls from 0.13 to −0.03. Its $\mathrm{MSSS}_{\mathrm{clim}}$ is less negative at every lead past ten days (mean −0.179 against −0.439) because it emits anomalies at $a \approx 0.54$ rather than 0.78, and a does not grow with lead; by the identity above, at $\mathrm{ACC} \approx 0.1$ the smaller amplitude is the cheaper error. Sea-surface temperature is again the channel that carries the skill (mean $\mathrm{MSSS}_{\mathrm{clim}}$ −0.006, positive at 33 of 73 leads, against −0.165 and 11 of 73 for the large head); sea-surface height is the sharpest one-step channel for both (0.87 and 0.84). Both heads beat raw persistence at nearly every lead and damped persistence at none, and the linear inverse model is more correlated than either from fifteen days on. The transport bands of the two heads have opposite signs (−0.025, +0.103, −0.244 against +0.107, −0.242, +0.163). The roll took 7,660 s against 77,147 s. One seed each; the 40.388M rung is not yet rolled.

Per channel. Only eight of the forty channels are scoreable. Sea-surface temperature holds correlation longest (0.76 at five days, 0.34 at 30 and at 90) and is the only channel whose $\mathrm{MSSS}_{\mathrm{clim}}$ stays positive beyond a few pentads; sea-surface height is sharpest at one step (0.84) and below 0.2 by a month; current speed and mixed-layer depth are one-step quantities; the four wind channels are noise past one step (Table 4).

channel	ACC 5 d	ACC 30 d	ACC 90 d	mean $MSSS_{clim}$	mean $MSSS_{damped}$
`sst`	0.759	0.343	0.337	−0.165	−0.191
`ssh`	0.838	0.252	0.113	−0.499	−0.562
`cur_speed`	0.604	0.131	0.052	−0.591	−0.606
`log_mld`	0.572	0.178	−0.009	−0.464	−0.469
`tau_x`	0.490	0.097	0.022	−0.383	−0.386
`tau_y`	0.235	0.039	−0.017	−0.426	−0.428
`tau_x_std`	0.222	0.134	0.104	−0.409	−0.410
`tau_y_std`	0.207	0.070	0.057	−0.465	−0.466

Table 4: Per-channel corridor scores of the rolled head.

Transport. The unpooled read-out of the rolled section states against RAPID gives $r = +0.107$ over 5–90 days (n=162), −0.242 over 95–180 (n=129), +0.163 over 185–365 (n=150). With about nine effective independent starts these are inside the noise of zero. A separate 36-month free run from a context ending 2004-12 — inside the training years, so not a skill measurement — correlates 0.636 with the true transport at pentad resolution and −0.13 on an 18-month low-pass at 61% amplitude: what the head tracks lives at high frequency, and the multi-year band is not measured by this roll.

Statistical power. The pool has 2,417 distinct end-times; consecutive windows share 143 of 144 frames, and every pixel is a correlated view of the same temporal patterns. With 206M parameters that is ∼85,000 parameters per pattern, and the training loss falling long after the held-out loss has turned is the expected consequence. For field-level scores the effective sample count is in the thousands and differences can be resolved; for the transport target (RAPID ∼20 years, anomalies decorrelating over months to years) it is of order ten, and a band correlation of ±0.2 cannot be. Every rolled number here is one seed; rolled skill at this cadence has not been replicated, and no two rolled numbers can yet be called different.

4 Related work

Observing the overturning. Continuous measurement of the Atlantic meridional overturning is two decades old. The RAPID-MOCHA-WBTS array at 26.5°N has measured the transport since April 2004 as the sum of the Florida Current cable, the Ekman transport from wind stress and the geostrophic interior from boundary density moorings [1]; the record reads 17.0 ± 2.8 Sv with a trend of −1.0 [−0.4, −1.6] Sv per decade that is marginally significant and dominated by a 2004–2010 drop and a 2010s pause [2, 3], with the climate signal carried by the deep and thermocline components rather than by the Gulf Stream or Ekman terms [2]. The Florida Current itself shows four decades of steady state once the cable is corrected for geomagnetic secular variation [4]. MOVE at 16°N monitors the deep western boundary since 2000 [5]; OSNAP measures the subpolar overturning since 2014 and places most of it east of Greenland [6, 7]; a 2026 synthesis of four western-boundary arrays reads a meridionally consistent decline in deep western transport between 16.5° and 42.5°N, partly compensated at the eastern boundary [8]. Whether the overturning weakened before 2004 is contested between sea-surface-temperature fingerprints [9], which read a ∼15% decline, and air–sea heat-flux fingerprints, which read none since the 1960s [10]. Our transport target is the RAPID series deseasonalised at monthly and pentad resolution, with MOVE as a second target, and our inputs carry no transport measurement.

Reconstructing transport from observables. The classical line regresses the 26.5°N transport on physically chosen observables: satellite sea level with the cable and Ekman terms [11], sea level through a thermal-wind argument [12], or boundary densities [13]; these reach r of 0.8–0.95 on 18-month low-passed series with in-sample calibration, and the one out-of-sample case holds out a single year. The machine-learning line trains on simulations: a convolutional network reading bottom pressure, sea level, temperature and wind strips reaches $r \approx 0.98$ inside a laminar 1° state estimate [14], the same task in eddy-resolving models explains 0.44–0.74 of the variance [15, 16], and a graph network on Argo profiles, wind stress and the boundary currents trained on VIKING20X recovers about 80% of the seasonal variance at 26.5°N with monthly errors under 1 Sv [16]; deep-learning reconstructions from surface fields trained on climate models extend an index back to 1900 [17]. Our read-out differs from all of these in protocol rather than in method: real RAPID data, monthly or pentad resolution with no low-pass filter, no transport among the inputs, and a year-blocked k-fold in which every year is held out in turn. On those terms a frozen embedding reads transport at $r = 0.67$ and a raw-pixel head at 0.66 (§2), which is where the eddying-model results place the difficulty of the task; the same read-outs low-passed at 18 months score 0.33–0.75 across codec seeds on nine effective points, which is why the report quotes unfiltered numbers.

Forecasting the overturning. Initialised decadal prediction systems show positive skill for subpolar ocean heat content at lead years 2–10 and for coastal sea level along the North American east coast at 3–5 years [18, 19], but the direct transport record is too short for the overturning itself to be verified at those leads; at 26.5°N the signal-to-noise ratio of the trend is not expected to exceed 2 before about 2040 [2]. The verification framework used here — mean-squared skill scores against climatology and persistence, anomaly correlation per lead, a damped-persistence reference — is the one established for interannual-to-decadal prediction [20]. The linear inverse model [21] is the standard empirical forecast against which such systems are judged, including as a benchmark for decadal forecasts of surface temperature [22]; the version in §3 fits it in the same standardised field the transformer is scored on. Statistical early-warning analyses of sea-surface-temperature fingerprints have placed a tipping of the overturning in the mid-21st century [23], but the estimate moves by a century or more with the choice of dataset and fingerprint [24], and physics-based indicators such as the freshwater transport at 34°S [25] are the alternative; none of this is a forecast at the pentad-to-seasonal leads measured here, where the question is whether any learned model adds to a linear operator.

Learned ocean and climate forecasting. Neural ocean forecasters at 1/4° to 1/12° and the global weather models they descend from [26, 27] operate at daily leads on full reanalysis fields and are cited for orientation; at monthly cadence, convolutional ENSO forecasts hold Niño3.4 correlation above 0.5 to about 17 months [28] and sea-surface-temperature anomaly fields are forecast to 18 months with recurrent networks [29]. Architecturally the codec is a masked autoencoder [30] over channel tokens, and the frozen-representation evaluation used throughout follows the argument that frozen backbones with shallow read-outs are the honest measure of a foundation model's representation [31]; the closest statement of the per-pixel embedding-field idea is terrestrial [32].

5 What the results support

For the current-state transport read-out the codec is at parity with the raw fields it compresses; its value, if any, is as a substrate that a transformer can attend over and roll forward, and on that use the transformer has learned what it learns inside two thousand steps at any width, rolls the same field at 7.6M parameters as at 206.7M, forecasts sea-surface temperature and height for days to weeks, does not beat a damped-persistence null at any lead, and is out-forecast beyond ten days by a linear inverse model with 200 modes fitted to the same field. The linear model is therefore the reference against which any further change to the forecaster is scored: what a learned model adds at leads of one to two pentads, and whether any change beats the linear model at three. The next measurements are the ones that bound this further: a retrieval baseline and a linear inverse model in embedding space through the identical battery, with block-bootstrap intervals; the early-stopped checkpoint and the 40.4M head rolled against the end states shown here; amplitude calibration fitted on training-year starts; and a codec trained on ≤ 2020 so that a terminal 2021–2024 test can be run once.

References

[1] G. McCarthy et al. (2015). Measuring the Atlantic meridional overturning circulation at 26°N. *Progress in Oceanography* 130.

[2] G. McCarthy et al. (2025). Signal and noise in the AMOC at 26°N. *Geophysical Research Letters* 52.

[3] S.-K. Lee et al. (2024). A pause in the weakening of the Atlantic meridional overturning circulation since the early 2010s. *Nature Communications* 15.

[4] D. Volkov et al. (2024). Florida Current transport observations reveal four decades of steady state. *Nature Communications* 15.

[5] U. Send, M. Lankhorst, T. Kanzow (2011). Observation of decadal change in the Atlantic meridional overturning circulation using 10 years of continuous transport data. *Geophysical Research Letters* 38.

[6] M. S. Lozier et al. (2019). A sea change in our view of overturning in the subpolar North Atlantic. *Science* 363.

[7] Y. Fu, M. S. Lozier et al. (2025). Characterizing the interannual variability of North Atlantic subpolar overturning. *Geophysical Research Letters* 52.

[8] X. Xing, S. Elipot, W. Johns, D. Smeed, B. Moat, J. Loder (2026). Meridionally consistent decline in the observed western boundary contribution to the AMOC. *Science Advances* 12.

[9] L. Caesar et al. (2018). Observed fingerprint of a weakening Atlantic Ocean overturning circulation. *Nature* 556.

[10] J. Terhaar, L. Vogt, N. Foukal (2025). Atlantic overturning inferred from air–sea heat fluxes indicates no decline since the 1960s. *Nature Communications* 16.

[11] E. Frajka-Williams (2015). Estimating the Atlantic overturning at 26°N using satellite altimetry and cable measurements. *Geophysical Research Letters* 42.

[12] A. Sanchez-Franks et al. (2021). A dynamically based method for estimating the AMOC at 26°N from satellite altimetry. *Ocean Science* 17.

[13] E. Worthington et al. (2021). A 30-year reconstruction of the Atlantic meridional overturning circulation shows no decline. *Ocean Science* 17.

[14] A. Solodoch et al. (2023). Machine learning-derived inference of the meridional overturning circulation from satellite-observable variables. *JAMES* 15.

[15] Y. Meng et al. (2024). Circumpolar transport and overturning strength inferred from satellite observables using deep learning. *JAMES* 16.

[16] E. Wölker, W. Rath, M. Renz, A. Biastoch (2025). Estimating the AMOC from Argo profiles with machine learning trained on ocean simulations. *Ocean Science* 21.

[17] S. Michel et al. (2025). Deep-learning reconstructions of the Atlantic meridional overturning circulation since 1900. *Environmental Research Letters* 20.

[18] S. Yeager et al. (2018). Predicting near-term changes in the Earth system: a large ensemble of initialized decadal prediction simulations using the Community Earth System Model. *BAMS* 99.

[19] T. Carmo-Costa et al. (2025). Decadal prediction skill of North Atlantic ocean heat content in CMIP6 systems. *Earth System Dynamics* 16.

[20] L. Goddard et al. (2013). A verification framework for interannual-to-decadal predictions experiments. *Climate Dynamics* 40.

[21] C. Penland, P. Sardeshmukh (1995). The optimal growth of tropical sea surface temperature anomalies. *Journal of Climate* 8.

[22] M. Newman (2013). An empirical benchmark for decadal forecasts of global surface temperature anomalies. *Journal of Climate* 26.

[23] P. Ditlevsen, S. Ditlevsen (2023). Warning of a forthcoming collapse of the Atlantic meridional overturning circulation. *Nature Communications* 14.

[24] M. Ben-Yami et al. (2024). Uncertainties too large to predict tipping times of major Earth system components from historical data. *Science Advances* 10.

[25] R. van Westen, M. Kliphuis, H. Dijkstra (2024). Physics-based early warning signal shows that AMOC is on tipping course. *Science Advances* 10.

[26] K. Bi et al. (2023). Accurate medium-range global weather forecasting with 3D neural networks. *Nature* 619.

[27] R. Lam et al. (2023). Learning skillful medium-range global weather forecasting. *Science* 382.

[28] Y.-G. Ham, J.-H. Kim, J.-J. Luo (2019). Deep learning for multi-year ENSO forecasts. *Nature* 573.

[29] J. Taylor, M. Feng (2022). A deep learning model for forecasting global monthly mean sea surface temperature anomalies. *Frontiers in Climate* 4.

[30] K. He et al. (2022). Masked autoencoders are scalable vision learners. *CVPR*.

[31] F. Lehmann et al. (2026). Frozen foundation-model backbones with shallow decoders as an evaluation standard. *JGR Machine Learning and Computation*.

[32] C. Brown, M. Kazmierski, V. Pasquarella et al. (2025). AlphaEarth Foundations: an embedding field model for accurate and efficient global mapping from sparse label data. arXiv:2507.22291.

A Data and code availability

Code: `https://github.com/blauewelt/earth` (codec trainer `ml/train.py`, stage-2 trainer `ml/temporal.py` and its JAX port `ml/jaxport/`, evaluator `ml/rollout_spatial.py`). Result bundles are on the repository's `ml-metrics` branch (`probes-<n>.json`; the rolls here are `probes-516.json` and `probes-520.json`, the linear inverse model `probes-527.json` (`ml/lim_baseline.py`), the codec probe ladder is in the bundles named in `ml/EXPERIMENTS.md`). Weights are on the `model-checkpoints-v1` release (`run-415__pixelmae.pt`; `head-weights-e059-200k-window-s0.pt` and the `e060a/e060b` heads named likewise); the tensor on the `data-cache-v1` release and the Hugging Face dataset `chfrank/earth-tensors`; the codec's embedding of it on `embed-cache-v1`. Stage-2 figures and tables are generated by `ml/paper/make_figs.py` from `ml/paper/data/report_data.json`, built by `ml/paper/extract_data.py` from those bundles and the training records; codec probe numbers are transcribed from the archived per-run bundles, with the source named beside each in the figure script.

Sources: GLORYS12 and the GREP ensemble (E.U. Copernicus Marine Service Information, used under the CMEMS licence with credit to the service); the Roemmich–Gilson Argo climatology (Scripps; Argo data are collected and made freely available by the International Argo Program and the national programs that contribute to it); NCEP/NCAR Reanalysis-1 (NOAA PSL); OISST v2.1 (NOAA NCEI); the RAPID-MOCHA-WBTS transport time series (NERC, NSF, NOAA); the MOVE transport time series (NOAA).